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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/02**

Paper 2 Theory

May/June 2004

1 hour 30 minutes

Candidates answer on the Question Paper.
Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in blue or black pen.
Do not use staples, paper clips, highlighters, glue or correction fluid.
You may use a calculator.

Sections A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer any **three** questions.
Write your answers on any lined pages and/or separate answer paper.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
A copy of the Periodic Table is printed on page 16.

This document consists of **15** printed pages and **1** lined page.

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Section A

Answer **all** the questions in this section in the spaces provided.

A1 Choose from the following substances to answer the questions below.

argon

calcium phosphate

ethene

lead(II) nitrate

methane

phosphorus oxide

potassium nitrate

sulphur dioxide

Each substance can be used once, more than once, or not at all.

Name a substance which,

(a) is a greenhouse gas produced by the decay of vegetable matter,

.....[1]

(b) contains **two** of the essential elements needed by plants,

.....[1]

(c) reacts with warm aqueous sodium hydroxide and aluminium powder to form a gas that turns moist red litmus blue,

.....[1]

(d) dissolves in water to form a solution which neutralises sodium hydroxide.

.....[1]

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A2 Two isotopes of potassium are $^{39}_{19}K$ and $^{40}_{19}K$.

(a) Complete the table about the number of particles found in one atom of each of these isotopes.

	protons	number of electrons	neutrons
$^{39}_{19}K$			
$^{40}_{19}K$			

[2]

(b) Potassium reacts with water as shown in the equation.

$$2K(s) + 2H_2O(l) \longrightarrow 2K^+(aq) + 2OH^-(aq) + H_2(g)$$

Describe what you would see when potassium reacts with water.

.....

.....

.....[2]

(c) A sample of 0.195 g of potassium was added to 500 cm^3 of cold water. When the reaction was finished, 100 cm^3 of 0.100 mol/dm^3 hydrochloric acid was added to form solution **X**.

(i) Calculate the number of moles of hydroxide ions formed when the potassium was added to water.

(ii) Calculate the number of moles of hydrogen ions in 100 cm^3 of 0.100 mol/dm^3 hydrochloric acid.

(iii) Give an ionic equation to represent the neutralisation reaction.

.....

(iv) Suggest a pH value for solution **X**.
Explain your answer.

.....

.....
[4]

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(d) Potassium oxide is an ionic solid.

Draw the electronic structure of both a potassium ion and an oxide ion.
Include the charge on each ion.

Potassium ion

Oxide ion

[2]

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A3 More than 60 000 plastic materials, or polymers, are in use.

The table gives some information about five important polymers.

polymer	density in kg/m^3	maximum useable temperature / °C	solubility in organic solvents
low density poly(ethene)	920	85	soluble above 80 °C
high density poly(ethene)	960	120	soluble above 80 °C
poly(phenylethene)	1050	65	soluble
poly(chloroethene)	1390	60	soluble
poly(propene)	900	150	insoluble

(a) Which polymer would be most suited for making a pipe to carry lubricating oil at 100 °C? Give **two** reasons for your answer.

answer .....

reasons .....

.....

.....[2]

(b) State **one** use for poly(ethene).

.....[1]

(c) Describe some of the problems of the disposal of waste polymers.

.....

.....

.....[2]

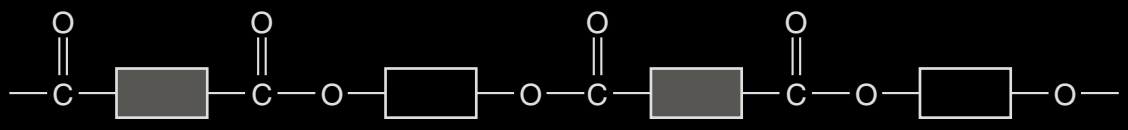
(d) Poly(propene) is made from the monomer propene.
Draw the structure of poly(propene).

[2]

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(e) *Terylene* is a condensation polymer.
The structure of *Terylene* is shown below.



(i) What is the name of the linkage shown in the structure of *Terylene*?

.....

(ii) Name a natural macromolecule that contains the same linkage as *Terylene*.

.....

[2]

(f) Draw the structure of a polyamide such as nylon.

[1]

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A4 The exhaust fumes from the internal combustion engines of motor vehicles contribute poor quality of air in many cities. The exhaust fumes contain atmospheric pollutants suc nitric oxide, NO, and carbon monoxide, CO.

(a) Nitric oxide, NO, is formed when oxygen and nitrogen from the air react in an internal combustion engine.

(i) Construct a balanced equation for this reaction.

.....

(ii) Explain why, in terms of collisions between particles, the rate of this reaction increases as the concentration of oxygen increases.

.....

.....

(iii) Explain why the rate of this reaction increases as the engine temperature increases.

.....

.....

[4]

(b) Explain how carbon monoxide is formed in an internal combustion engine.

.....

.....[1]

(c) Nitric oxide and carbon monoxide in the exhaust gases react together in the catalytic converter of a motor vehicle.

(i) Write a balanced equation for this reaction.

.....

(ii) Explain why the catalyst should be in the form of a powder supported on a mesh.

.....

.....

.....

[3]

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A5 Electrolysis is the decomposition of a liquid by the passage of an electrical current.

(a) Aqueous copper(II) sulphate contains the following ions, Cu^{2+}, H^+, OH^- and SO_4^{2-}.
Aqueous copper(II) sulphate can be electrolysed using inert electrodes.

The electrode reactions are represented below.

cathode $Cu^{2+} + 2e^- \longrightarrow Cu$
anode $4OH^- \longrightarrow O_2 + 2H_2O + 4e^-$

(i) Explain why copper, **not** hydrogen, is formed at the cathode.

.....

.....

(ii) Explain why the formation of oxygen at the anode is an example of oxidation.

.....

.....

(iii) The electrolysis of aqueous copper(II) sulphate using copper electrodes has a different anode reaction.
Give the equation for the electrode reaction at the anode.

.....
[3]

(b) Molten lead(II) bromide decomposes when an electric current is passed through it.

(i) Explain why solid lead(II) bromide will not conduct electricity but molten lead(II) bromide will.

.....

.....

.....

(ii) Construct the equations for the two electrode reactions.

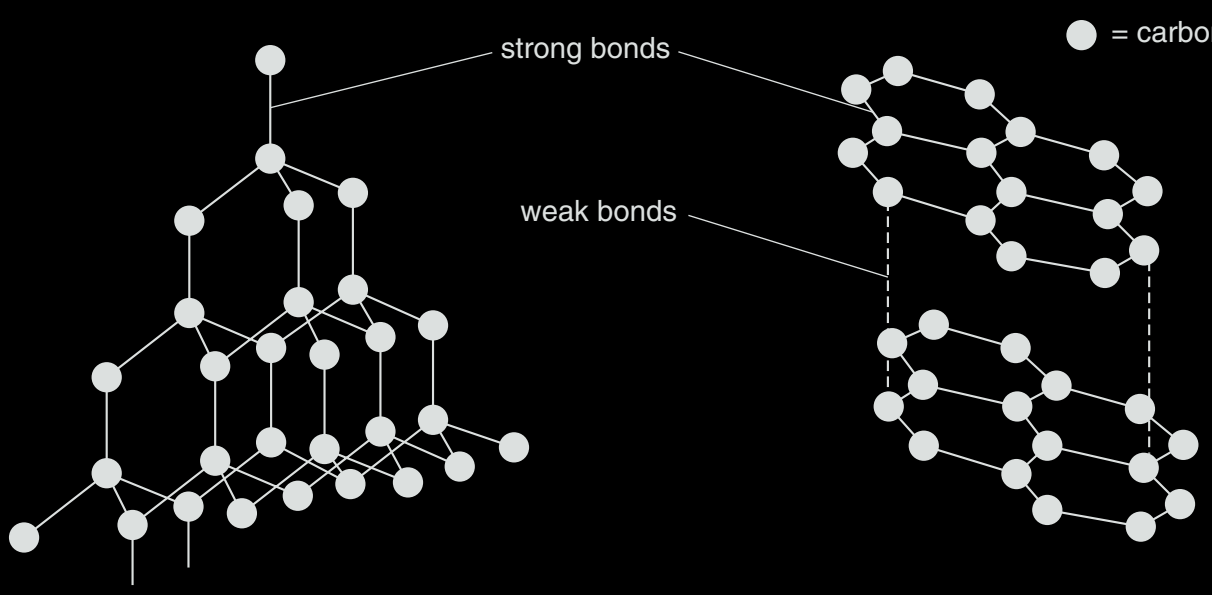
cathode .....

anode .....
[4]

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A6 The structures of diamond and graphite are drawn below.



diamond graphite

(a) Name the type of strong bond shown on the diagram.

.....[1]

(b) Diamond has a melting point of about 3700 °C and graphite has a melting point of about 3300 °C.

(i) Explain why both diamond and graphite have very high melting points.

.....

.....

.....

(ii) Suggest why the melting point of graphite is lower than that of diamond.

.....

.....
[3]

(c) Compare the electrical conductivity of diamond and graphite.
Explain your answer.

.....

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.....[2]

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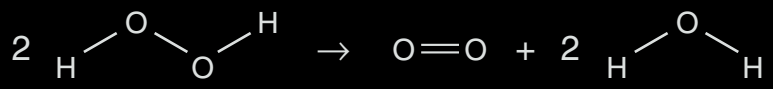
Section B

Answer **three** questions from this section.

B7 Aqueous hydrogen peroxide is used to sterilise contact lenses.

At room temperature aqueous hydrogen peroxide decomposes very slowly to form water and oxygen.

The decomposition can be represented by the equation below.



$\Delta H = -206\,\text{kJ/mol}$

(a) Explain why this reaction is exothermic in terms of the energy changes that take place during bond breaking and bond making. [2]

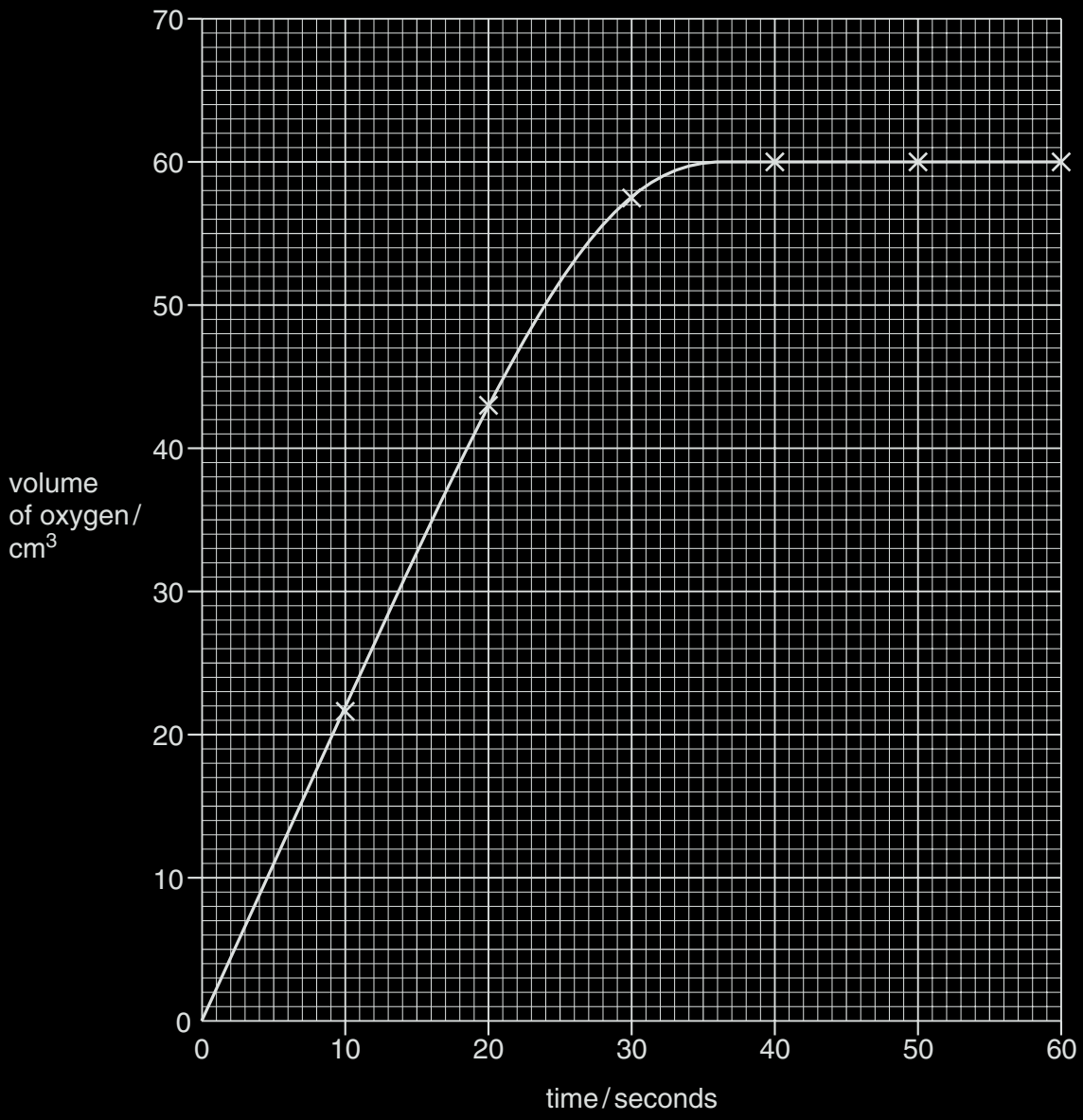
(b) Draw the energy profile diagram for the decomposition of hydrogen peroxide.
Label on the diagram the activation energy and the enthalpy change. [3]

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(c) Manganese(IV) oxide catalyses the decomposition of aqueous hydrogen perox... an experiment 50.0 cm^3 of aqueous hydrogen peroxide was mixed with 0.50 ... manganese(IV) oxide. The total volume of oxygen formed was measured every ... seconds.

The results of the experiment are shown in the graph.



(i) After how many seconds did the decomposition of hydrogen peroxide finish?

(ii) How many moles of oxygen were produced at the end of the decomposition? [At room temperature and pressure one mole of oxygen occupies 24000 cm^3.]

(iii) Use your answer to **(ii)** to calculate the concentration, in mol/dm^3, of the 50.0 cm^3 of aqueous hydrogen peroxide used in the experiment.

[5]

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B8 Nickel is a transition element. It is manufactured in a four-stage process from ni sulphide, NiS.

- Stage 1 – nickel(II) sulphide is heated in air to form nickel(II) oxide and sulphur dioxid
- Stage 2 – nickel(II) oxide is heated with carbon to give impure nickel.
- Stage 3 – impure nickel is reacted with carbon monoxide to make nickel tetracarbonyl, $Ni(CO)_4$.
- Stage 4 – nickel tetracarbonyl is decomposed to give pure nickel.

(a) **(i)** Construct the balanced equation for the reaction in stage 1.

(ii) Calculate the mass of sulphur dioxide that is formed when 182 kg of nickel sulphide is heated in air. [3]

(b) Nickel tetracarbonyl is a liquid with a boiling point of 43 °C.
Suggest, with a reason, the type of bonding in nickel tetracarbonyl. [2]

(c) Suggest **one** possible environmental consequence of the manufacture of nickel. [1]

(d) Give an example of the use of nickel as a catalyst. [1]

(e) In an experiment, small amounts of three metals were added to three aqueous metal nitrate solutions.
The results are shown in the table.

	aqueous zinc nitrate $Zn(NO_3)_2$	aqueous nickel(II) nitrate, $Ni(NO_3)_2$	aqueous copper(II) nitrate, $Cu(NO_3)_2$
zinc	no reaction	green solution went colourless and zinc coated with a silver solid	blue solution went colourless and zinc coated with a pink solid
nickel		no reaction	
copper	no reaction	no reaction	no reaction

Predict the observations when nickel is added to separate solutions of zinc nitrate and copper(II) nitrate.
Write an ionic equation for **one** of the reactions that takes place. [3]

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B9 Ethene is an important starting material for the production of chemicals such as e ethanoic acid and ethane-1,2-diol. Ethene, C_2H_4, is manufactured by the cracking of chain hydrocarbons such as dodecane, $C_{12}H_{26}$.

(a) Construct an equation to show the cracking of dodecane to make ethene. [1]

(b) Draw a 'dot and cross' diagram for ethene. You only need to draw the valence (outer shell) electrons. [1]

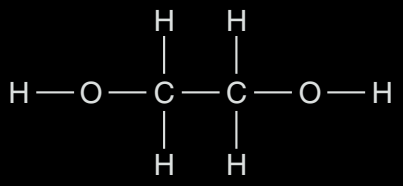
(c) Ethene can also be converted into a compound that contains carbon, hydrogen and oxygen. A sample of the compound was analysed and found to contain 0.72 g of carbon, 0.18 g of hydrogen and 0.96 g of oxygen.
Show that the empirical formula of the compound is CH_3O. [3]

(d) Describe how ethene can be converted industrially into ethanol. [2]

(e) Ethanol reacts with hot acidified potassium dichromate(VI) to form ethanoic acid.

(i) Describe the colour change that occurs during this reaction and draw the structure of ethanoic acid.

(ii) Ethane-1,2-diol has the structure drawn below.



Suggest the structure of the product of the reaction between ethane-1,2-diol and hot acidified potassium dichromate(VI).

[3]

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B10 The table below shows some of the ores of iron.

ore	formula
haematite	Fe_2O_3
magnetite	Fe_3O_4
siderite	$FeCO_3$

(a) Which ore in the table contains the greatest percentage by mass of iron? Explain your answer. [2]

(b) Give the equations for the **redox** reactions taking place in the extraction of iron from haematite.
In each case state which substance is oxidised and which is reduced. [4]

(c) Iron is malleable. Describe how this property can be explained in terms of its structure. [2]

(d) State and explain how the properties of iron can be changed by the addition of carbon. [2]

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DATA SHEET
The Periodic Table of the Elements

Group																
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium	226 Ra Radius 88	227 Ac Actinium 89 †															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

8-71 Lanthanoid series
90-103 Actinoid series

ey

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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